

Screens — Helix Connector (appliance UI)

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** the complete screen inventory and per-screen UX specification of **Helix Connector** — the HelixVPN appliance/network-side agent UI (Flutter, flavor: `HelixFlavor.connector`, capabilities `{Capability.tunnel, Capability.advertise, Capability.localAcl}`) [03-client-core-and-ui.md §6]. A Connector is the device that *advertises a subnet into the overlay* — it joins one or more private LANs to the HelixVPN network so authorized clients can reach LAN hosts (default-deny, C4).

It defines, screen by screen: purpose, layout (compact-first appliance form factor, responsive up to desktop config), the `helix_design` components used, every state (loading / empty / error / live), navigation, interactions, accessibility, and the **light + dark** rendering — each backed by an ASCII wireframe.

SPEC-ONLY. It describes *what each screen is and how it behaves* — not the shipping connector build. Original HelixVPN UX design work; the control-plane data each screen surfaces is owned by the cited control-plane docs.

Headless-first note (load-bearing). The Connector ships **both** a headless daemon entrypoint (the same Rust core, no UI — primary on appliances) **and** this **optional** Flutter config UI [03-client-core-and-ui.md §6, D-CLIENT-4]. On a true headless appliance there is no screen at all; this UI is the **web-managed** / **local-attended** surface for the same daemon. That headless $\rightleftharpoons$ UI duality is surfaced as decision **D-CONN-0** below.

Boundary with sibling docs. Owns: the Connector screen set + per-screen layout/state/interaction/a11y. **Consumes:** colour roles + connection / feedback semantics + per-app **teal** accent [color-system.md §3/§6]; token tiers + spacing/radius/breakpoint scales [design-tokens.md §6]; the signature components (ConnectButton, StatusChip, NetworkTile, ShieldIndicator) [component-library.md, sibling]; the OpenDesign engine [opendesign-foundation.md]; the **Console** counterpart that manages the same data centrally [screens-console.md, sibling]; the **Client** screens [screens-client.md, sibling]. The **data** is owned by the control-plane [02-control-plane.md] + [v03-control-plane/svc-registry.md], [.../svc-ipam.md], [.../svc-identity.md], [.../svc-coordinator.md]; the FFI status enum + shim by [03-client-core-and-ui.md §3/§4].

Evidence base. [CP §N] = final/02-control-plane.md; [CLIENT §N] = final/03-client-core-and-ui.md; [REG §N]/[IPAM §N]/[COORD §N] = the matching final/v03-control-plane/svc-*.md; [COLOR §N] = final/v10-design/color-system.md; [DT §N] = final/v10-design/design-tokens.md; [FFI §N] = final/03-client-core-and-ui.md §3 (the TunnelStatus/AdvertiseResult surface). Claims not grounded in the evidence base or in this document's own original design choices are tagged UNVERIFIED per §11.4.6 — never fabricated.

Table of contents

- 0. What the Connector is — and the principles that shape its screens
 - 1. The Connector shell (form factor, responsive law)
 - 2. Screen inventory + navigation map
 - 3. Enroll / onboarding (enroll token, key generation)
 - 4. Connection / health status (the home screen)
 - 5. Advertised routes (the subnets it exposes)
 - 6. Route conflicts (overlapping-CIDR surfacing)
 - 7. Settings (appliance mode, kill-switch typically Off)
 - 8. Cross-screen states, a11y & light/dark contract
 - 9. Surfaced decisions & cross-doc contracts
 - Sources verified
-

0. What the Connector is — and the principles that shape its screens

A **Connector** is a device with `kind = connector` [CP §2.2, REG §2] that **advertises CIDRs** into the overlay so authorized clients can reach the private LAN behind it — the HelixVPN "1 user → N networks" differentiator [CP §3.1]. It runs the *same* Rust core as the Client, in `CoreMode.Connector / advertise mode` [CLIENT §6], and is typically an always-on appliance (a small box at a site, warehouse-gw, office-gw). Its UI is intentionally **narrow**: get enrolled, prove it is connected and healthy, declare which subnets it exposes, and surface any conflicts — nothing more. Central management (policy, who-may-reach-it) lives in the **Console** [screens-console.md]; the Connector UI is the *local* face.

Five principles govern every Connector screen:

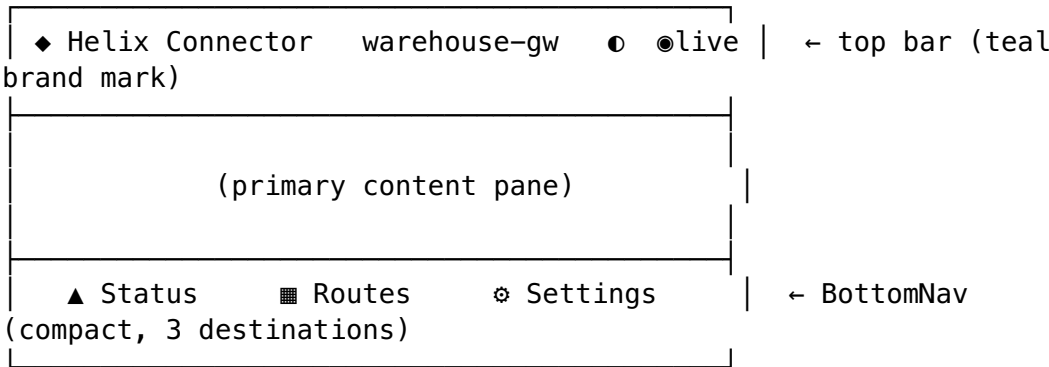
1. **Status legibility above all.** Like the Client, the Connector's emotional centre is "am I up and serving my subnets?" — the home screen is a big, unambiguous health surface driven by the core's `statusStream` [FFI §3.2, CLIENT §0.1 CI2]. The UI is a **pure function of that stream** — never paints "up" on intent alone.
2. **Infrastructure accent = teal, safety colours unchanged.** The Connector's brand-overridable accent is `teal.*` [COLOR §6] — it reads as a network/ infrastructure device, distinct from the Client's indigo and the Console's violet. The **connection-state and feedback semantics never change** (D-COLOR-1): "connected" is the same green everywhere.
3. **Declare subnets safely, conflict-aware.** The core advertise action returns `AdvertiseResult { accepted, conflicts }` [FFI §3.1]; the UI shows accepted vs conflicting CIDRs honestly and never claims a conflicting prefix is live.
4. **Appliance defaults, not consumer defaults.** A Connector is a server, not a personal device — so the kill-switch (which would sever the LAN it serves) is **Off by default**; turning it on is a deliberate, explained choice (§7, D-CONN-1).
5. **Headless-first, UI-optional.** The same daemon runs with **no screen** on a true appliance (D-CLIENT-4). This UI is the optional local/web-managed surface; it must degrade gracefully to "the daemon is the source of truth" and never imply the UI is required for the Connector to function.

Honest boundary (§11.4.6). This document specifies layout, state, and interaction. It does **not** re-specify the FFI/REST payloads (owned by [FFI]/[CP §8]) nor the token hex/contrast (owned by [COLOR]/[DT]). A value not grounded in those is marked UNVERIFIED.

1. The Connector shell (form factor, responsive law)

The Connector UI runs on whatever screen the appliance offers — often a small local display or, more commonly, a **browser pointed at the appliance** (local management) — so it is **compact-first** but renders the full responsive ladder [CLIENT §7.3, DT §6.6]:

Breakpoint [DT §6.6]	Width	Shell layout
compact	< 600	BottomNavigationBar (Status · Routes · Settings) + single pane — the appliance default
medium	600–1024	NavigationRail + master/detail (routes list route editor)
expanded	> 1024	extended NavigationRail + two-pane (e.g. routes conflict detail) — desktop config

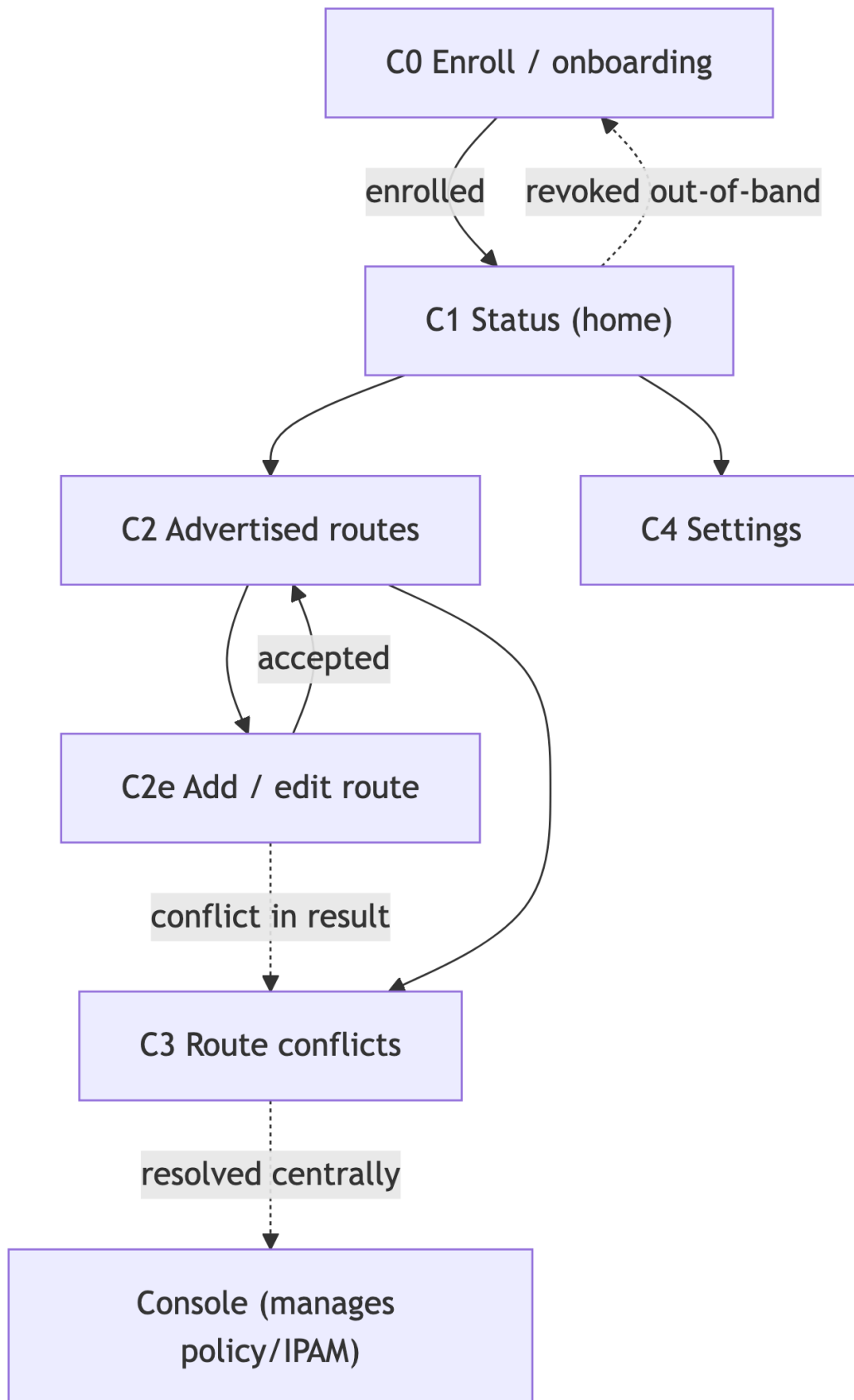


- **Top bar:** teal brand mark, the connector's name (`site_name` [CP §2.2]), a theme toggle (☾ system/light/dark), and a **live indicator** (● green stream healthy / amber reconnecting / red down — reuses the connection-state palette [COLOR §3]).
- **Three primary destinations** keep the appliance UI minimal: **Status** (home), **Routes** (advertised subnets, with the conflicts sub-view), and **Settings**. Onboarding (Enroll) is a pre-enrollment flow that precedes the shell.

2. Screen inventory + navigation map

#	Screen	Route	Primary data source	Components
C0	Enroll / onboarding	/enroll		

#	Screen	Route	Primary data source	Components
			identity enroll-token + local WG keygen [CP §9.2, FFI §3]	token field/QR scan, keygen, progress steps
C1	Connection / health status (home)	/	statusStream [FFI §3.2] + presence [REG §5]	ConnectButton/ health hero, StatusChip, serving summary
C2	Advertised routes	/routes	advertised_prefixes + advertise() [REG §6, FFI §3.1]	route table, add/edit CIDR, enable toggle
C2e	Route — add / edit	/routes/ edit	local form → advertise()	CIDR input, validation, accepted/ conflict result
C3	Route conflicts	/routes/ conflicts	route.conflict.detected [REG §6.2, IPAM §6]	conflict list, 4via6 explanation, guidance
C4	Settings	/settings	local config + appliance mode	mode, kill- switch (Off default), transport, identity, danger zone



The Connector is a **near-linear** flow: enroll once → live on the Status home → declare routes → handle conflicts. Deep policy/IPAM resolution is delegated to the Console (the conflict view *explains* and links, it does not resolve centrally itself). An out-of-band device. revoke from the Console [CP §9.3] drops the Connector back to an enroll-required state.

3. Enroll / onboarding (enroll token, key generation)

Purpose. First-run: turn a single-use **enroll token** (minted by an operator in the Console [CP §9.2]) into an enrolled connector identity — the device **generates its WG keypair locally and the private key never leaves it** (C6 [CP §9.2]); only the 32-byte public key is registered. The flow mirrors the enrollment sequence in [CP §9.2] from the device side.

Layout (stepper, compact-first).

◆ Set up this Connector

●

○

○

○

token keys name done

Step 1 · Enroll token

Paste the token from your Console, or scan its QR code.

HLX-.....

[Scan QR]

Control-plane URL

https://cp.acme.example

[Continue ►]

← step indicator

← token input

Steps. (1) **Token + control-plane URL** — paste/scan the single-use token; (2) **Key generation** — the UI shows "generating WireGuard keypair on this device — the private key never leaves here" with a brief progress state (the keygen is local, C6); (3) **Name / site** — confirm the connector's site_name [CP §2.2]; (4) **Done** — Enroll(token, wg_pubkey, os, kind=CONNECTOR) is sent [CP §4]; on success the device receives {device_id, overlay_ip, device_cert, gateway} [CP §9.2] and the UI advances to the Status home.

Step 2 · Generating keys

🔑 Creating WireGuard keypair...

The private key is generated on this

device and never transmitted (C6).	
 done in a moment	

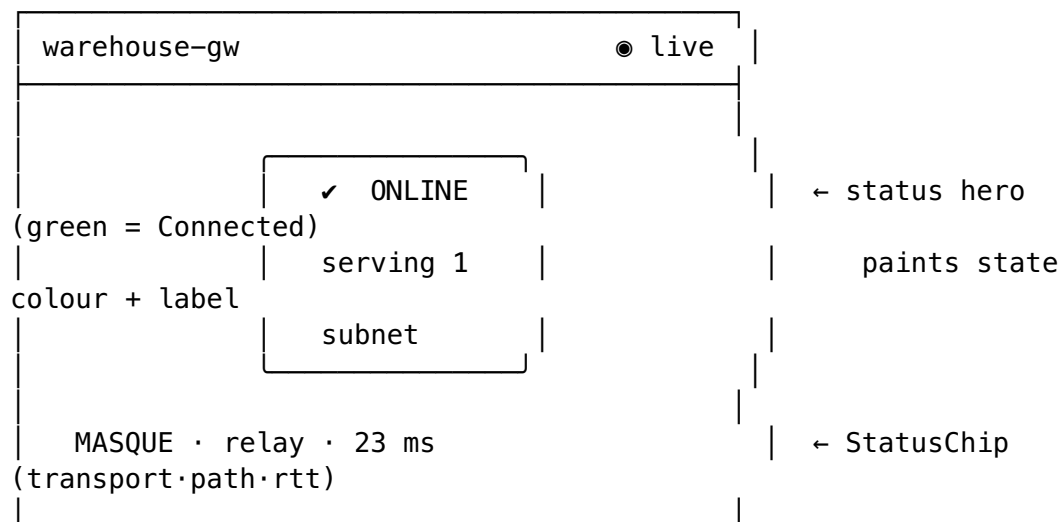
States. *Loading* — per-step progress; keygen shows honest progress, never a fake instant "done". *Empty* — first-run default (no token yet). *Error* — an invalid/expired/already-consumed token returns a precise error ("this token was already used — mint a new one in the Console", `feedback.error`); an unreachable control-plane URL is a distinct error with retry. *Success* — a brief confirmation (overlay IP assigned `fd7a:helix:...` [IPAM §1]) then auto-advance to C1.

Interactions / a11y. Token field is the default focus; QR scan uses the device camera (with a manual-paste fallback for headless/no-camera appliances). The token is **never logged** (§11.4.10). Full keyboard nav; step indicator has SR labels. **Light/dark:** the stepper card is `surface.raised`; [Continue] is `action.primary` (teal #0F766E light / #2DD4BF dark — AA-proven [COLOR §6]); the key-privacy note is `feedback.info/text.secondary`.

4. Connection / health status (the home screen)

Purpose. The Connector's home and emotional centre: **is this appliance up, connected to the gateway, and serving its subnets?** It is a pure function of the core's `statusStream` [FFI §3.2, CI2] — the 7-variant `TunnelStatus` (`Disconnected` · `Connecting` · `Handshaking` · `Connected{direct|relay}` · `Reconnecting` · `Down` · `Danger`) drives the hero colour + label via the connection-state palette [COLOR §3], **always with a glyph + text** (CI5).

Layout (status hero + serving summary).



Serving		← NetworkTile per advertised CIDR
■ 10.10.0.0/24	✓ advertised · enabled	
Overlay IP	fd7a:helix:a1b2::5	
Gateway	gw.acme.example ● reachable	
ⓘ This appliance exposes its LAN subnets to authorized clients only (default-deny)		

Per-state hero rendering (the 7-variant mapping, [COLOR §3.1]):

TunnelStatus [FFI §3.2]	Hero label	Hero colour family [COLOR §3]
Disconnected	"Offline — not connected (by choice)"	grey
Connecting / Handshaking	"Connecting..."	amber
Connected{direct}	"Online — serving · direct"	green (pure)
Connected{relay}	"Online — serving · relay"	teal-green sub-shade
Reconnecting	"Reconnecting..." (amber pulse , reduce-motion → static)	amber
Down{reason}	"Dropped — <honest reason>, retrying"	orange
Danger{kind}	"Danger — <leak/killswitch> (z-top, overrides)"	red

Components. The **status hero** (a Connector-tuned ConnectButton/health surface — for an always-on appliance it reads as a *status* surface more than a tap target, but tapping toggles connect/disconnect for local control); a **StatusChip** showing transport · path · RTT (MASQUE · relay · 23ms) live from Connected{transport,path,rtt} [FFI §3.2]; a **serving summary** — one NetworkTile per enabled advertised CIDR [CLIENT §7.2], the overlay IP, and the gateway reachability dot.

States. *Loading* — hero in a neutral "connecting..." skeleton until the first statusStream event (never green before the stream confirms it). *Empty* — enrolled but **no routes advertised yet** ⇒ hero shows online but the serving summary shows "no subnets advertised — add one in Routes" with a CTA to C2. *Error/Down* — Down{reason} paints orange with the **honest reason** [FFI §3.2, §11.4.6] and a retry; gateway-unreachable is distinguished from core-down. *Danger* — Danger{kind} overrides everything with the red banner at

z-top [COLOR §3.2/§5] (e.g. a leak/kill-switch trip). *Revoked* — an out-of-band device. revoke flips the hero to "revoked — re-enrollment required" and routes to C0.

A11y / light-dark. The hero state is **announced** (SR), never colour-only (CI5); the amber Reconnecting pulse degrades to static under reduce-motion (never strobes, [COLOR §3.2], §11.4.107). Light/dark via the connection-state palette (every state AA-proven both themes [COLOR §4.3/§4.4]); the teal accent appears only on brand/action chrome, never on a safety signal.

5. Advertised routes (the subnets it exposes)

Purpose. The Connector's defining function: declare which LAN **CIDRs** this appliance advertises into the overlay [REG §6]. Add / edit / enable / disable prefixes; each change runs through the core `advertise(cidrs)` action that returns `AdvertiseResult { accepted, conflicts }` [FFI §3.1] (the same code path as a Console-side prefix edit [REG §6.1]).

Layout (route table + add).

Advertised routes (2)		[+ Add CIDR]	
CIDR	ENABLED	STATUS	
10.10.0.0/24	✓	✓	advertised ← accepted, live
10.20.0.0/24	✓	⚠	conflict ← overlaps another connector → C3
10.30.0.0/24	○	disabled	← toggled off locally
2 advertised · 1 conflict · 1 disabled			
⚠ 1 conflict – review		→ tap → C3	

Add / edit (C2e).

< Routes Add CIDR

CIDR

10.40.0.0/24

a CIDR on blur

ⓘ Advertise a private subnet this appliance can route to. Reachability is governed by your Console policy (default-deny).

[Cancel] [Add ▶]

← validated as

RESULT (from advertise()) ✓ accepted: 10.40.0.0/24 (or) △ conflict: overlaps office-gw → C3

Components. A **route table** (one row per advertised_prefixes entry [REG §6] with CIDR, enable toggle, per-row status: advertised / conflict / disabled); an **add/edit form** validating CIDR syntax locally before calling advertise(); the **AdvertiseResult** rendered honestly — accepted CIDRs flip to "✓ advertised", conflicts flip to "△ conflict" and link to C3 (never shown as live [FFI §3.1]).

Interactions. Add a CIDR → advertise() → result shown inline; toggle enabled (writes the prefix's enabled flag [CP §2.2]); edit/remove with a §11.4.66 confirm for removal (it withdraws a subnet authorized clients may be using). A conflicting result does **not** block the local declaration but is clearly marked non-live and routed to C3 — resolution is central (Console/IPAM [REG §6.3]).

States. *Loading* — table skeleton. *Empty* — "no subnets advertised yet — add the LAN CIDR this appliance should expose" with the primary CTA. *Error* — a failed advertise() shows the honest core error + retry; an invalid CIDR is caught at the form before the call. *Conflict present* — conflict rows are amber with a count badge linking to C3.

A11y / light-dark. Status cells pair colour with a glyph + text (✓/△, never colour-only). The CIDR field uses font.primitive.family.mono [DT §4]. Light: table on surface.raised, accepted = feedback.success, conflict = feedback.warning; dark mirrors with the same roles.

6. Route conflicts (overlapping-CIDR surfacing)

Purpose. Surface and **explain** overlapping-CIDR conflicts — when this Connector advertises a subnet that another connector in the tenant also advertises (route.conflict.detected [REG §6.2, IPAM §6]). The defining HelixVPN problem ("two home/lab LANs both 192.168.1.0/24" [CP §3.1]) shown from the appliance side. Conflicts are **advisory, never blocking** [REG §6.3]; resolution is central (Console/IPAM) — this screen makes the situation legible and points the right way.

Layout.

< Routes Route conflicts (1)

△ 10.20.0.0/24 This appliance (warehouse-gw, site-1) advertises a subnet that overlaps: office-gw (site-2) How HelixVPN resolves this (4via6): Each site gets a unique IPv6 mapping so clients can reach the right host: warehouse → fd7a:helix:a1b2:0001::/96 office → fd7a:helix:a1b2:0002::/96 ① Conflicts are advisory and never block. Resolve centrally in the Console / IPAM, or keep both via 4via6 disambiguation. [open in Console]

Components. A **conflict list** (each overlapping-CIDR pairing with the other connector's name + site-id); a **4via6 explanation** spelling out the per-site / 96 IPv6 mappings that disambiguate the collision [IPAM §4.3, CP §3.1 D4]; an **advisory note** + a deep-link to the Console's IPAM/conflict view where central resolution happens [screens-console.md §10].

States. *Loading* — list skeleton. *Empty* — "no conflicts — all advertised subnets are unique" (the healthy default). *Error* — inline retry over last-known. The screen is **read + explain + link**; it does not itself perform central resolution (that is a Console/operator action per §11.4.66 [REG §6.3]).

A11y / light-dark. Conflict headers carry a △ glyph + text; the 4via6 mappings are monospace. Light/dark via surface/text/feedback roles; the conflict emphasis is amber in both themes; no colour-only signalling (CI5).

7. Settings (appliance mode, kill-switch typically Off)

Purpose. Local appliance configuration. Crucially, the Connector's **defaults differ from the Client's** because it is a server, not a personal device — the kill-switch is **Off by default** (turning it on would sever the very LAN the appliance serves on a control-channel blip).

Layout (sectioned).

Settings
► Appliance

Mode	● headless-managed ○ local	← D-CONN-0
Name/site	warehouse-gw	
Auto-start	✓ (start on boot)	
▶ Protection		
Kill-switch	○ Off (default)	← D-CONN-1 — Off
for appliances		
① On a connector, kill-switch would cut the LAN it serves if control drops. Leave Off unless you understand this.		
DNS protection	○ Off	
(DAITA / post-quantum — client shields, not typically a connector concern)		
▶ Transport		
Escalation ladder	plain-udp → lwo → masque-h3 (from CP)	
▶ Appearance		
theme	● system ○ light ○ dark	
▶ Identity		
overlay	fd7a:helix:a1b2::5	
cert valid · auto-renews		
[Re-enroll]		
▶ Danger zone		
Leave network (de-enroll)		← confirmed, irreversible

Components & key decisions.

- **Appliance mode** (D-CONN-0): *headless-managed* (the daemon is primary, this UI is a window onto it) vs *local* (UI-attended). Auto-start-on-boot toggle.
- **Protection** — the ShieldIndicator-class controls [CLIENT §7.2] wired to the core Shields [FFI §3.1], but with **appliance defaults: kill-switch Off** (D-CONN-1) with an explicit explanation of why a connector should usually leave it off (it serves a LAN; cutting it on a blip is rarely desired). DAITA / post-quantum are surfaced as client-oriented shields and de-emphasised here.
- **Transport** — the escalation ladder the control plane ships (TransportPolicy.order [CP §4]); shown read-mostly (the Console owns the default; local override is allow_user_override-gated [CP §4]).
- **Identity** — overlay IP + cert status (auto-renewing over the control channel [CP §9.3]) + a re-enroll action.
- **Danger zone** — "Leave network (de-enroll)" — irreversible, §11.4.66 double-confirmed (it withdraws the appliance and its subnets).

States. *Loading* — section skeletons. *Error* — per-section inline. *Kill-switch toggle* — turning it **On** raises a §11.4.66 confirm explaining the LAN-severing consequence (it is not a silent flip). *Revoked* — identity section shows revoked + re-enroll guidance.

A11y / light-dark. Every toggle is keyboard-operable + labelled; the kill-switch warning is `feedback.warning` (amber both themes); danger-zone uses `feedback.error` + a typed confirm. Standard surface/text roles, light + dark.

8. Cross-screen states, a11y & light/dark contract

These hold for **every** Connector screen (so each section above states only its specifics):

Concern	Contract	Source
Status truth	the UI is a pure function of the core <code>statusStream</code> ; never paints "up/serving" on intent alone	CI2 [CLIENT §0.1], [FFI §3.2]
Loading	skeleton placeholders, never a green-before-confirmed hero; the status hero waits for the first stream event	original UX, §11.4.6
Empty	purposeful empty states with the next action (enroll, add a CIDR) — never an ambiguous blank	original UX
Error / Down	the honest reason from <code>Down{reason}</code> / core error, an inline retry; gateway-unreachable distinguished from core-down	§11.4.6, [FFI §3.2]
Danger	<code>Danger{kind}</code> (leak / kill-switch tripped) overrides everything, red, at z-top, never occluded	[COLOR §3.2/§5], [FFI §3.2]
Confirm	every consequential action (remove route, enable kill-switch, de-enroll) is confirmed via the interactive mechanism	§11.4.66
Colour ≠ only signal	every state dot/badge pairs colour with a glyph + text; the Reconnecting pulse degrades to static under reduce-motion	CI5 [CLIENT §0.1], [COLOR §0/§3.2]
No overlap / no label-overlay	≥ <code>space.scale.2</code> (8 px) between coloured regions; overlays use <code>overlay.scrim</code> ; danger at z-top	§11.4.162, [COLOR §5]
Light + dark	every screen renders in both, all roles AA-proven both themes; teal accent is the only per-app difference	§11.4.162, [COLOR §2/§4/§6]

Concern	Contract	Source
Headless honesty	the UI never implies it is required; on a headless appliance the daemon is the source of truth and the UI is an optional window	D-CLIENT-4 [CLIENT §6]

The teal light/dark accent (the Connector's only brand difference).

Role	Light	Dark	Proof [COLOR §6]
action.primary / accent	teal.700 #0F766E	teal.400 #2DD4BF	text 5.47 light / 9.47 dark — AA both

Everything else — surfaces, text, borders, **connection-state**, **feedback** — is the identical shared semantic set used by Client and Console (D-COLOR-1); the Connector never recolours a safety signal. (Note: the teal accent and the Connected{relay} sub-shade are deliberately the same hue family — but the relay state is still distinguished from the brand accent by the paired glyph + "relay" label, never colour alone, [COLOR §3.2].)

9. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-CONN-0	Headless-first: the Connector daemon runs with no UI on a true appliance (primary); this Flutter UI is the optional local/web-managed surface for the same core. The UI degrades to "daemon is source of truth" and never implies it is required.	decided [CLIENT §6, D-CLIENT-4]
D-CONN-1	The kill-switch is Off by default on a Connector (a server serving a LAN), unlike the Client (On-favoured). Enabling it is a §11.4.66-confirmed choice with an explicit LAN-severing warning.	decided (recommended)
D-CONN-2	Conflicts are surfaced + explained (4via6) locally but resolved centrally (Console/IPAM); the Connector UI links out, it does not perform central resolution.	decided [REG §6.3]
D-CONN-3		decided

id	Decision / contract	Status
	Minimal three-destination shell (Status · Routes · Settings) — the appliance UI stays narrow; policy / who-may-reach-it lives in the Console.	
C-CONN-A (consumes)	The TunnelStatus/AdvertiseResult/Shields FFI surface owned by [FFI §3]; control-plane data + payloads by [CP §8] + [REG]/[IPAM].	contract
C-CONN-B (consumes)	Colour roles + teal accent owned by [COLOR]; token scales/breakpoints by [DT]; signature components by [component-library.md].	contract
C-CONN-C (provides)	This screen set + per-screen states are the spec the visual-regression + golden suite asserts against (light+dark, every breakpoint, §11.4.162).	contract
U-CONN-1 UNVERIFIED	The exact local-management transport for the web-managed UI (does the browser UI talk to the local daemon over a local HTTP/IPC socket, or only via the control plane?) is pinned by the shim/daemon spec [CLIENT §5]; stated here as "web-managed", binding UNVERIFIED.	open
U-CONN-2 UNVERIFIED	Whether the Connector UI exposes a local-ACL/site-restriction editor (the flavor carries Capability.localAcl [CLIENT §6]) as its own screen vs folding it into Settings — surfaced; the MVP folds local-ACL into Settings, a dedicated screen is UNVERIFIED/Phase-2.	open

Sources verified

- **Connector flavor** ({`tunnel`, `advertise`, `localAcl`}), **headless daemon** + optional Flutter config UI (**D-CLIENT-4**), **capability model**, `runHelixApp`, the signature components (`ConnectButton/StatusChip/NetworkTile/ShieldIndicator`), **responsive law**, **ally** (UI = **pure function of status stream**, **colour ≠ only signal**, **reduce-motion**) — `final/03-client-core-and-ui.md` §6 (flavors/capabilities),

- §7 (design system + responsive + ally), §3 (the FFI TunnelStatus/ AdvertiseResult/ Shields surface), §4 (TunnelPlatform shim), §0.1 (CI2/CI5), §14 (D-CLIENT-4 headless+UI) (read 2026-06-25).
- **Connector control-plane data** — final/02-control-plane.md: §2.2 (devices/connectors/advertised_prefixes DDL, kind=connector), §3 (IPAM ULA /48 + 4via6, the colliding-subnet problem D4), §4 (Coordinator proto: Enroll, AdvertisePrefixes, AdvertiseResponse{conflicts}, TransportPolicy), §9 (enrollment: local WG keygen, private key never leaves C6; cert auto-renew; revoke < 1 s) (read 2026-06-25).
 - **Per-service screen data** — final/v03-control-plane/: svc-registry.md §2/§5/§6 (connector/device model, presence, prefix advertisement + intra-tenant overlap conflict detection, advisory-not-blocking semantics); svc-ipam.md §1/§4/§6 (ULA address plan, per-connector site-id, 4via6 route derivation, conflict lifecycle); svc-coordinator.md §4 (event reactions: connector.attached, connector.prefixes.changed); svc-identity.md (enroll-token issue/consume) (read 2026-06-25).
 - **Colour roles, the Connector teal accent + its AA contrast proofs, the 7-variant connection-state palette (the status-hero mapping), reduce-motion pulse rule, no-overlap / no-label-overlay rule, light+dark mandate** — final/v10-design/color-system.md §3 (connection-state palette + per-state rationale), §4 (contrast proofs), §5 (no-overlap rule), §6 (per-app accent: Connector = teal) (sibling, this wave).
 - **Token tiers, spacing/radius/breakpoint scales (the responsive breakpoints + surfaces the layouts use), motion (connectPulse/statexfade), the mono font** — final/v10-design/design-tokens.md §6 (scales/motion), §4 (font), §1 (tiers) (sibling, this wave).
 - **OpenDesign engine that emits the tokens** — final/v10-design/opendesign-foundation.md (sibling, this wave; §11.4.162).
 - **Signature-component contracts + the Console (central management) / Client counterpart screens** — final/v10-design/component-library.md, .../screens-console.md, .../screens-client.md (siblings; screens-console + the component library land in this same Volume-10 wave — cross-referenced, not duplicated).
 - Items marked UNVERIFIED (U-CONN-1 local-management transport, U-CONN-2 dedicated local-ACL screen vs Settings-fold) are pending their named decision/ contract spec per §11.4.6 — **not** asserted as fact.
 - **Layout, per-screen state machines, interaction flows, wireframes, the appliance-default decisions (kill-switch Off, headless-first, minimal shell) — NO external source needed — original HelixVPN UX design work** (the screen compositions are owned by this document; they render the cited control-plane / FFI data through the cited design-system tokens/components).